

Week 5 – Inverse Kinematics

ELEC0129 Introduction to Robotics

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Schedule

Legend:

A: MPEB 6.02 Teaching Lab

B: Roberts 7.04 PC Lab

Week	Recorded, uploaded by Friday of previous week	Workshop, Mondays 11am-1pm and/or Wednesdays 9am-11am	Notes
1	(Scenario Week)		
2	Lec: Intro; Spatial description	A: Workshop: Offline programming	Half class Mon, half class Wed
3	Lec & Tut: Spatial description	A: Workshop: Build robot	
4	Lec & Tut: Forward kinematics	A: Workshop: Forward kinematics	
5	Lec & Tut: Inverse kinematics	B: Workshop: Offline programming	Half class Mon, half class Wed
RW	(Reading Week)		
6	(Scenario Week)		
7	Lec & Tut: Jacobians	A: Workshop: Inverse kinematics	
8	Lec: Trajectory Planning	A: Workshop: Trajectory planning	
9	Lec & Tut: Dynamics	A: Workshop: Trajectory planning	
10	Lec & Tut: Control	A: Workshop: Pick-and-place demo	

Content

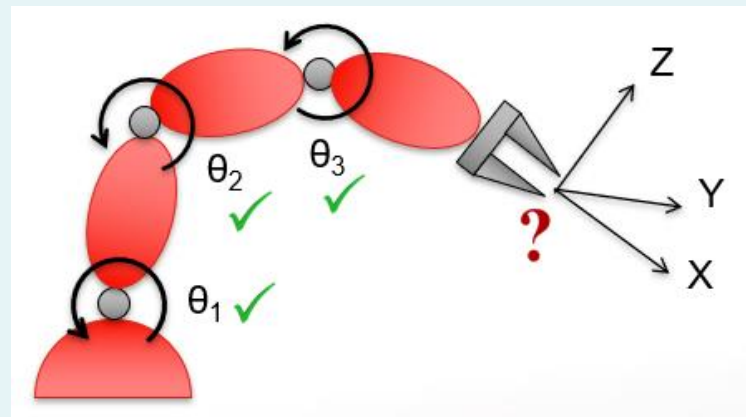
- Introduction to Inverse Kinematics
- Existence and Uniqueness of Solution
- Geometric Solution
- Algebraic Solution
- Another Example
- Tutorial Questions

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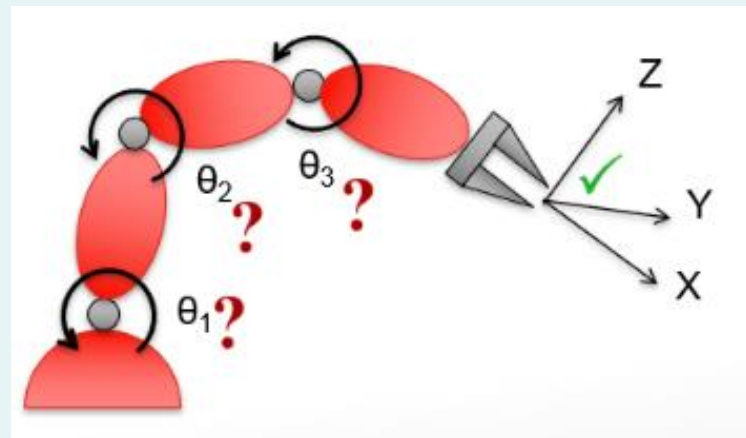
Introduction (1)

- In the previous lectures, we studied the **forward kinematics** problem:
 - Given the **joint space** parameters (angles for revolute joints, or offsets for prismatic joints),
 - What is the position and orientation of the end-effector in **Cartesian space**?



Introduction (2)

- Today, we will look at the **inverse kinematics problem** (much more difficult!):
 - Given the **desired position and orientation** of the tool in Cartesian space,
 - What is the **set of joint angles** that is required to achieve the desired outcome?



Introduction (3)

- For example, consider a 6-link robot.
- The **homogeneous transform** from base to link 6 is:

$${}^0_6T = \begin{bmatrix} f_{11}(q_1, \dots, q_6) & f_{12}(q_1, \dots, q_6) & f_{13}(q_1, \dots, q_6) & f_{14}(q_1, \dots, q_6) \\ f_{21}(q_1, \dots, q_6) & f_{22}(q_1, \dots, q_6) & f_{23}(q_1, \dots, q_6) & f_{24}(q_1, \dots, q_6) \\ f_{31}(q_1, \dots, q_6) & f_{32}(q_1, \dots, q_6) & f_{33}(q_1, \dots, q_6) & f_{34}(q_1, \dots, q_6) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Assume we want to **place the last frame** at:

$$\begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Introduction (4)

- We therefore need to solve the simultaneous equations to obtain 6 joint parameters q_1 to q_6 .

$$\begin{bmatrix} f_{11}(q_1, \dots, q_6) & f_{12}(q_1, \dots, q_6) & f_{13}(q_1, \dots, q_6) & f_{14}(q_1, \dots, q_6) \\ f_{21}(q_1, \dots, q_6) & f_{22}(q_1, \dots, q_6) & f_{23}(q_1, \dots, q_6) & f_{24}(q_1, \dots, q_6) \\ f_{31}(q_1, \dots, q_6) & f_{32}(q_1, \dots, q_6) & f_{33}(q_1, \dots, q_6) & f_{34}(q_1, \dots, q_6) \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- There are 12 non-trivial values in the matrix on the right hand side.
- From the 9 values related to rotation, only 3 are independent.
- And we have 3 values related to position.
- Therefore, there are altogether 6 independent equations.

Introduction (5)

- 6 equations and 6 unknowns → Can be solved!
 - However, it is not always easy to get the answers...
-

- Note: p_x, p_y, p_z in the previous two slides can also be the position of **end-effector** instead of last frame.
 - We just need to make sure that the functions f_{14}, f_{24}, f_{34} are for **end-effector** rather than the last frame.
 - f_{14}, f_{24}, f_{34} for end-effector can be obtained via ${}^0P = {}^0T \cdot {}^nP$.
 - **Assumption**: orientation of end-effector = orientation of last frame.

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- Introduction to Inverse Kinematics
- **Existence and Uniqueness of Solution**
- Geometric Solution
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Existence & Uniqueness of Solution

- The equations are nonlinear:

$$\begin{bmatrix} f_{11}(q_1, \dots, q_6) & f_{12}(q_1, \dots, q_6) & f_{13}(q_1, \dots, q_6) & f_{14}(q_1, \dots, q_6) \\ f_{21}(q_1, \dots, q_6) & f_{22}(q_1, \dots, q_6) & f_{23}(q_1, \dots, q_6) & f_{24}(q_1, \dots, q_6) \\ f_{31}(q_1, \dots, q_6) & f_{32}(q_1, \dots, q_6) & f_{33}(q_1, \dots, q_6) & f_{34}(q_1, \dots, q_6) \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

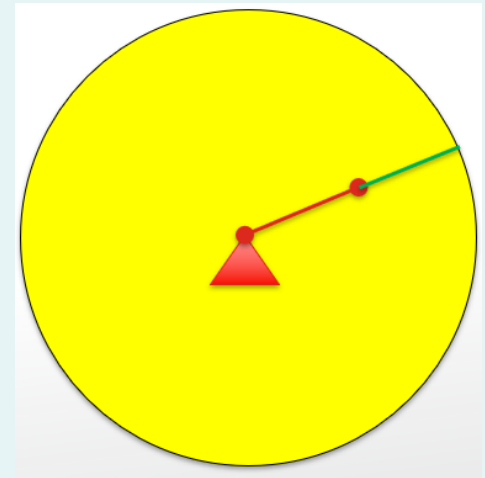
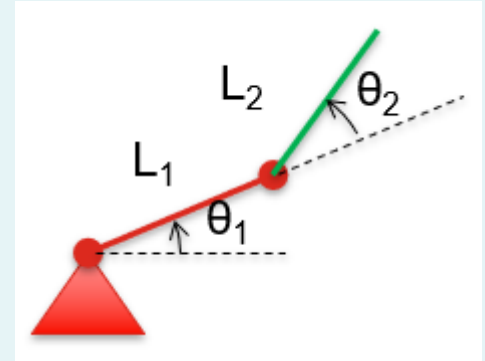
- Thus, two fundamental questions need to be answered:
 - Existence of Solutions
 - Uniqueness / Multiplicities of Solutions

Existence of Solution (1)

- Let's discuss the first question first.
- The **answer** is as follows:
 - If the target point lies within the **manipulator's workspace**,
 - then there **exists at least one set of solution** to the joint angles.
- But what is workspace?

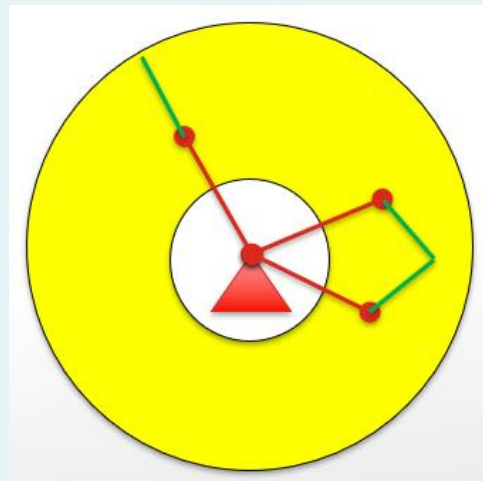
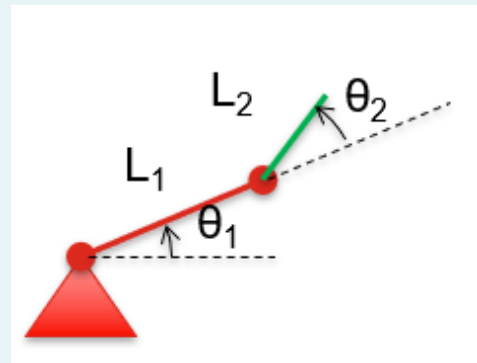
Existence of Solution (2)

- Roughly speaking, workspace is the **volume of space** which the **end-effector** can reach.
- E.g. Two-link robot with $L_1 = L_2$.
- Workspace: Disc of radius $L_1 + L_2$.



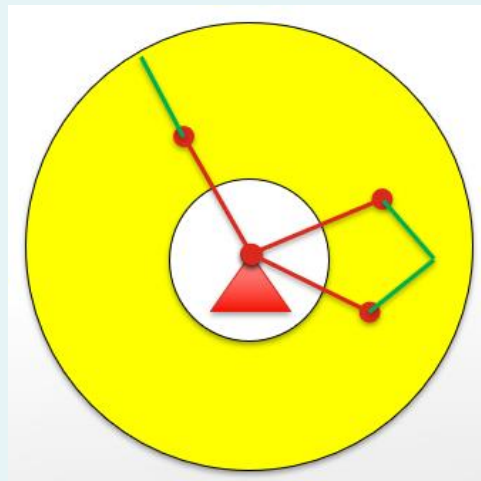
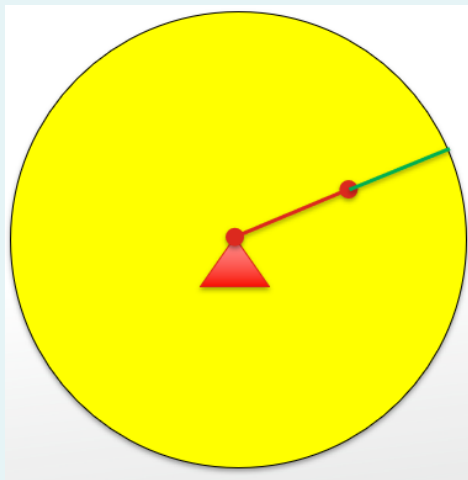
Existence of Solution (3)

- Another example: Two-link robot with $L_1 \neq L_2$.
- Workspace: Ring of outer radius $L_1 + L_2$ and inner radius $|L_1 - L_2|$.



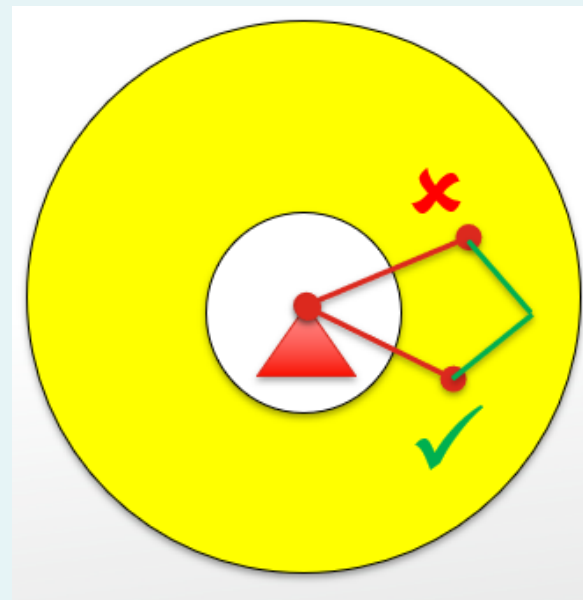
Existence of Solution (4)

- Note:
 - Usually > 1 orientations for points within workspace.
 - Only 1 orientation for points on boundary.



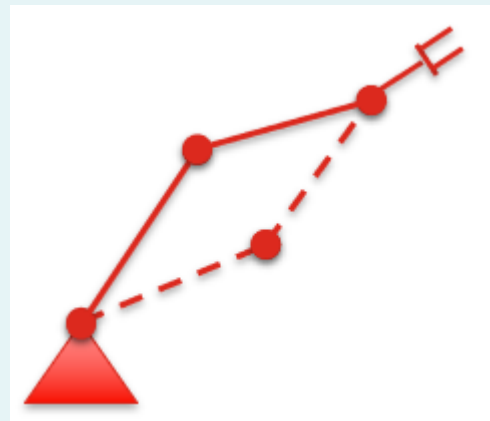
Existence of Solution (5)

- Note: We have assumed that the joints can rotate 360 degrees.
- In most robots, the **joint angles are actually limited**. Then:
 - Either the workspace is smaller; or
 - Number of possible orientations is reduced.
- E.g. Same example as before but θ_2 limited to 0 and 180 degrees.
 - Workspace volume is the same.
 - But only one orientation per point.



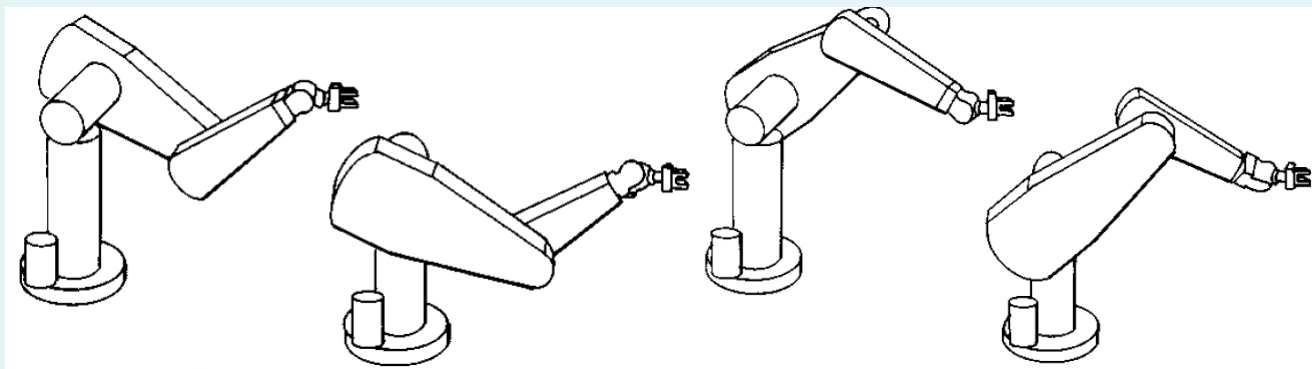
Multiplicities of Solutions (1)

- Next, we shall discuss about the second question, i.e. the **uniqueness / multiplicities of solutions**.
- E.g. 3-link RRR Manipulator.
 - **2 possible solutions**, to achieve one particular end-effector position and orientation.



Multiplicities of Solutions (2)

- E.g. PUMA robot.
 - 8 possible solutions for one position and orientation.
 - The first 4 are shown:



- The other 4 are similar to above except that:

$$\theta'_4 = \theta_4 + 180, \theta'_5 = -\theta_5, \theta'_6 = \theta_6 + 180$$

Multiplicities of Solutions (3)

- General notes:
 - A manipulator needs **at least 6-DOF** to place the end-effector at an **arbitrary position** with an **arbitrary orientation** in space.
 - If manipulator has **> 6-DOF**, then there are **infinite number of solutions**. This kind of manipulator is called “**redundant manipulator**”.

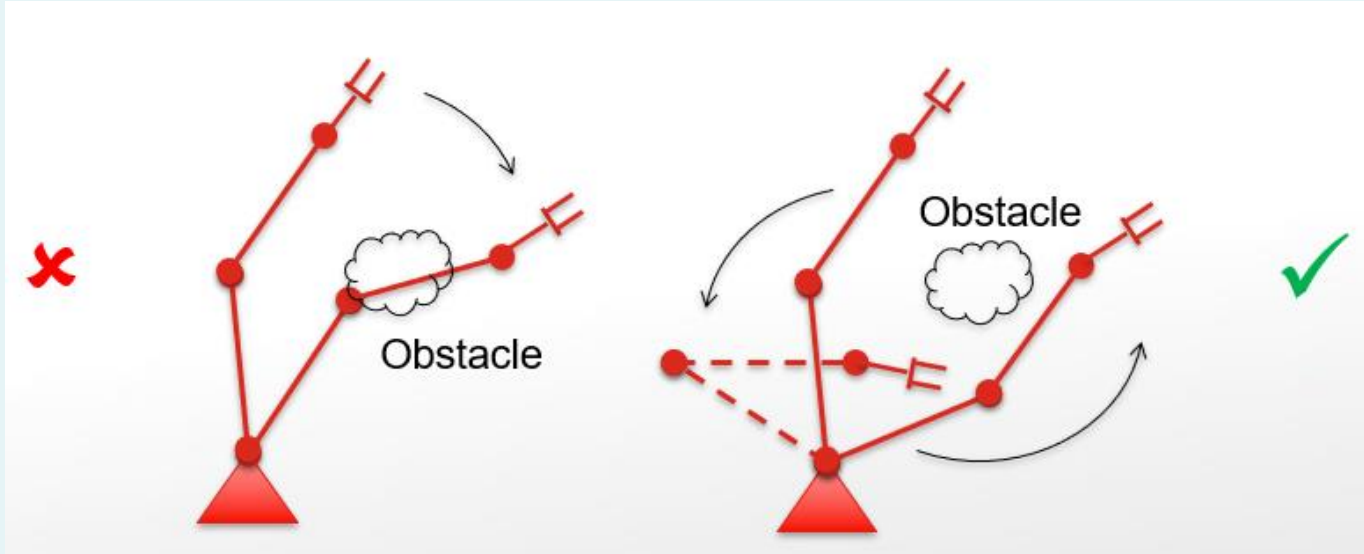


Elephant Trunk
Robotic Arm

https://en.wikipedia.org/wiki/Snake-arm_robot#/media/File:ElephantArmRobot.jpg

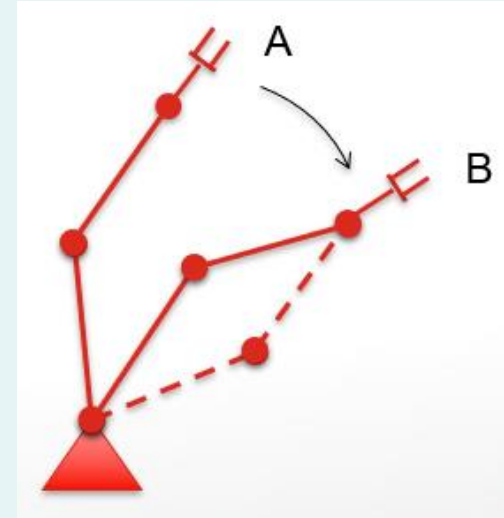
Multiplicities of Solutions (4)

- Advantage of multiplicities of solutions:
 - Collision avoidance.



Multiplicities of Solutions (5)

- Disadvantage of multiplicities of solutions:
 - The system has to be able to choose one out of many solutions!
 - If there are no obstacles, then a reasonable choice is to choose the “closest” solution.
 - E.g. to move from A to B.
 - The upper configuration is chosen because the joints move less.



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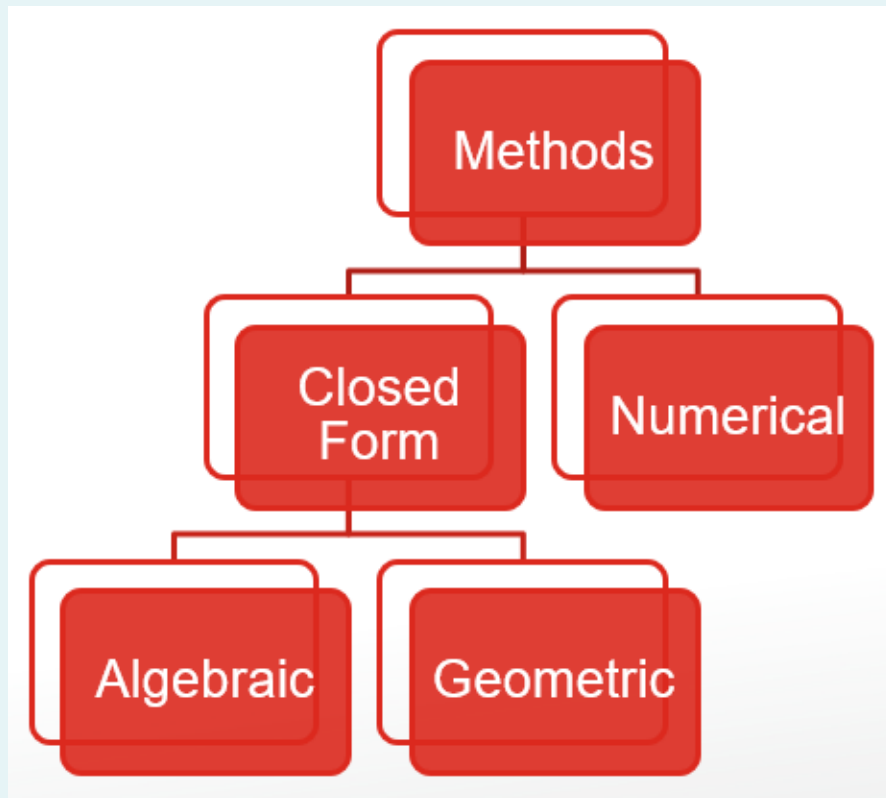
Reminder

- We want to solve for 6 joint parameters q_1 to q_6 , given desired orientation and position of last frame (or end-effector).

$$\begin{bmatrix} f_{11}(q_1, \dots, q_6) & f_{12}(q_1, \dots, q_6) & f_{13}(q_1, \dots, q_6) & f_{14}(q_1, \dots, q_6) \\ f_{21}(q_1, \dots, q_6) & f_{22}(q_1, \dots, q_6) & f_{23}(q_1, \dots, q_6) & f_{24}(q_1, \dots, q_6) \\ f_{31}(q_1, \dots, q_6) & f_{32}(q_1, \dots, q_6) & f_{33}(q_1, \dots, q_6) & f_{34}(q_1, \dots, q_6) \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

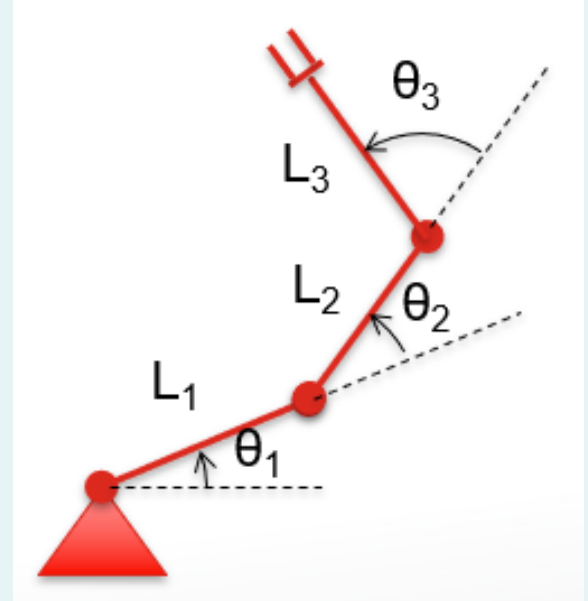
Methods of Solutions

- We will only look at the **closed-form** solutions.
- Let's discuss **geometric solutions** first.



Geometric Solution (1)

- Example: 3 link RRR robot.
- The desired position (x, y) and orientation (ϕ) of last frame given.



Geometric Solution (2)

- Using cosine rule:

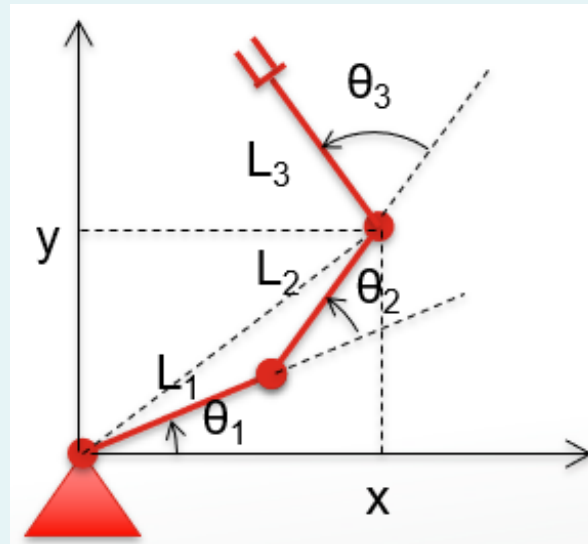
$$\begin{aligned}x^2 + y^2 &= L_1^2 + L_2^2 - 2L_1L_2 \cos(180 - \theta_2) \\&= L_1^2 + L_2^2 + 2L_1L_2 \cos(\theta_2)\end{aligned}$$

- This gives:

$$\theta_2 = \arccos\left(\frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}\right)$$

- Using symmetry for “elbow-up” case gives

$$\theta_2' = -\theta_2$$



Geometric Solution (3)

- To solve for θ_1 , note that:

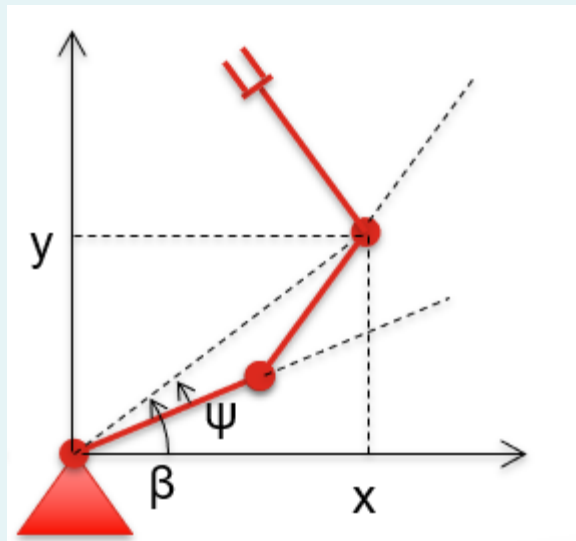
$$\beta = \text{atan2}(y, x)$$
$$\cos \psi = \frac{L_1^2 + x^2 + y^2 - L_2^2}{2L_1\sqrt{x^2 + y^2}}$$

- Then,

$$\theta_1 = \beta - \psi$$

- Using symmetry for “elbow-up” case,

$$\theta'_1 = \beta + \psi$$



Geometric Solution (4)

- Finally, θ_3 can be solved easily, because the sum of joint angles = final orientation:

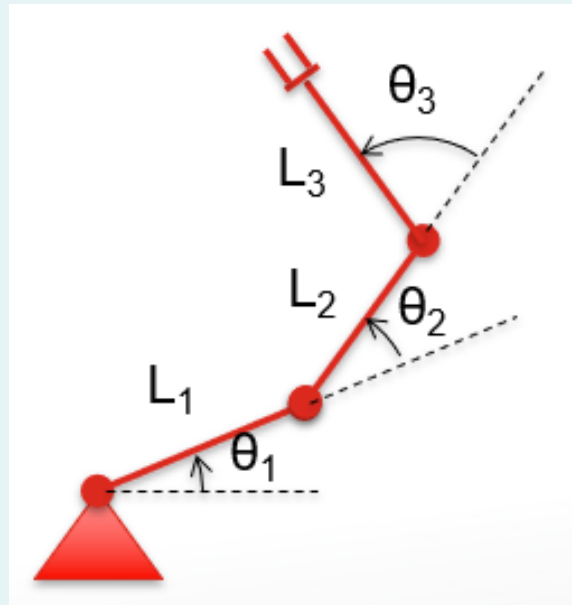
$$\theta_1 + \theta_2 + \theta_3 = \phi$$

- Giving:

$$\theta_3 = \phi - \theta_1 - \theta_2$$

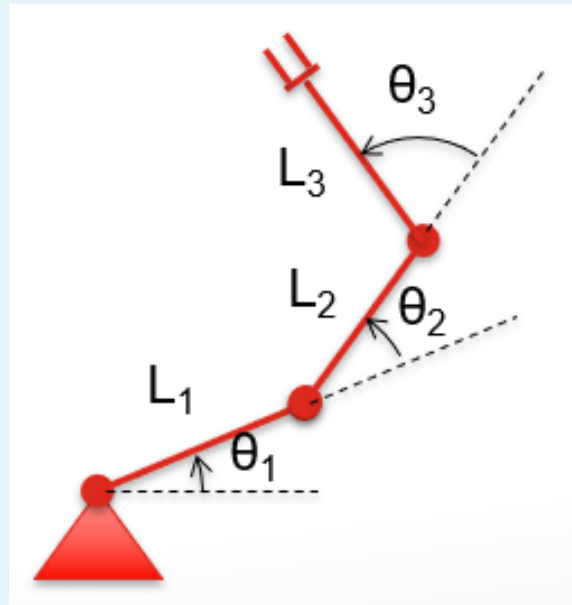
- For “elbow up” configuration,

$$\theta'_3 = \phi - \theta'_1 - \theta'_2$$



Geometric Solution (5)

- Note: There is **no general algorithm** to the inverse kinematics problem.
- We can only obtain the solutions on a **case-by-case** basis.
- The method of solution for the 3 link RRR robot will not be applicable to other robots.



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Algebraic Solution (1)

- The geometric approach may not be easy if the robot is non-planar.
- The algebraic approach attempts to solve the simultaneous equations more directly.

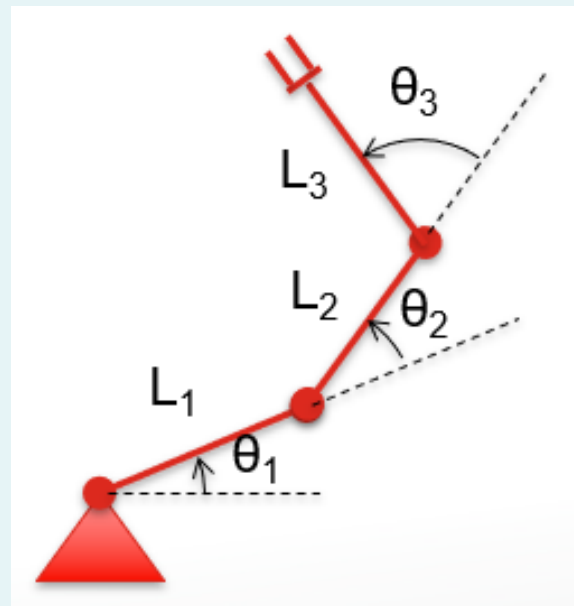
Algebraic Solution (2)

- Nevertheless, there is still **no general algorithm** to solve the inverse kinematics problem.
- Thus the solutions have to be calculated on a **case-by-case** basis.
- That said, **two techniques** are generally quite useful:
 - Square and sum.
 - Trigonometric substitution.

Algebraic Solution (3)

- Example: 3 link RRR robot.
- The **general** homogeneous transformation from base to frame {3} is:

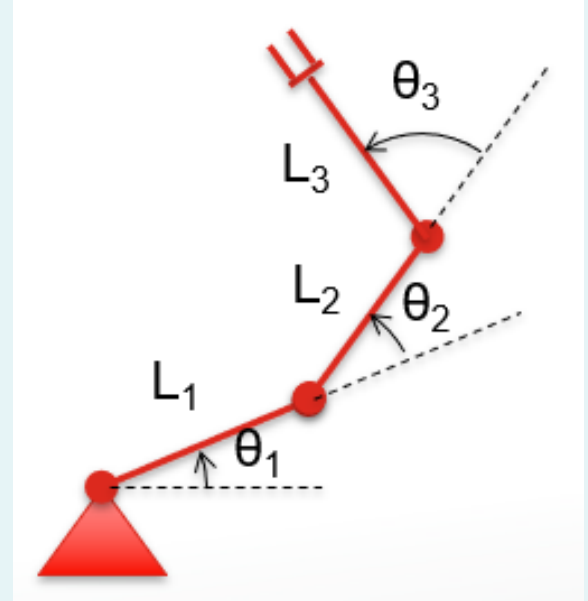
$${}^0_3T = \begin{bmatrix} c\theta_{123} & -s\theta_{123} & 0 & L_2c\theta_{12} + L_1c\theta_1 \\ s\theta_{123} & c\theta_{123} & 0 & L_2s\theta_{12} + L_1s\theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Algebraic Solution (4)

- Assume the **desired** orientation and position for the last frame are:

$$\begin{bmatrix} c\phi & -s\phi & 0 & x \\ s\phi & c\phi & 0 & y \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Algebraic Solution (5)

- Comparing the two matrices, we observe that:

$$c\phi = c\theta_{123} = \cos(\theta_1 + \theta_2 + \theta_3)$$

$$s\phi = s\theta_{123} = \sin(\theta_1 + \theta_2 + \theta_3)$$

$$x = L_1 c_1 + L_2 c_{12}$$

$$y = L_1 s_1 + L_2 s_{12}$$

- Aim: Solve for $\theta_1, \theta_2, \theta_3$ from ϕ, x, y .

Algebraic Solution (6)

- Apply the “square and sum” technique.

$$\begin{aligned}x^2 &= L_1^2 c_1^2 + 2L_1 L_2 c_1 c_{12} + L_2^2 c_{12}^2 \\y^2 &= L_1^2 s_1^2 + 2L_1 L_2 s_1 s_{12} + L_2^2 s_{12}^2 \\x^2 + y^2 &= L_1^2 + 2L_1 L_2 (c_1 c_{12} + s_1 s_{12}) + L_2^2\end{aligned}$$

- From trigonometry identity:

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

- Therefore:

$$\begin{aligned}c_1 c_{12} + s_1 s_{12} &= \cos(\theta_1 - (\theta_1 + \theta_2)) \\&= \cos(-\theta_2) \\&= \cos(\theta_2)\end{aligned}$$

Algebraic Solution (7)

- Hence:

$$\begin{aligned}x^2 + y^2 &= L_1^2 + 2L_1L_2(c_1c_{12} + s_1s_{12}) + L_2^2 \\ &= L_1^2 + 2L_1L_2c_2 + L_2^2\end{aligned}$$

- We can now solve for θ_2 :

$$c_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}$$

- This value must be between -1 and 1 for a solution to exist.

Algebraic Solution (8)

- The answer for θ_2 is not complete yet. It doesn't show **multiplicities of solutions**.
- To do that, we also write:

$$s_2 = \pm \sqrt{1 - c_2^2}$$

Where the choice of sign corresponds to elbow up or down solutions.

- We then **calculate** θ_2 as:

$$\theta_2 = \text{atan2}(s_2, c_2)$$

Algebraic Solution (9)

- Next, we shall try to solve for θ_1 .

$$\begin{aligned}x &= L_1 c_1 + L_2 c_{12} \\&= L_1 c_1 + L_2 c_1 c_2 - L_2 s_1 s_2 \\&= (L_1 + L_2 c_2) c_1 - (L_2 s_2) s_1 \\y &= L_1 s_1 + L_2 s_{12} \\&= L_1 s_1 + L_2 s_1 c_2 + L_2 c_1 s_2 \\&= (L_1 + L_2 c_2) s_1 + (L_2 s_2) c_1\end{aligned}$$

- Introducing:

$$\begin{aligned}K_1 &= L_1 + L_2 c_2 \\K_2 &= L_2 s_2\end{aligned}$$

- These are known because L_1, L_2, c_2, s_2 are all known.

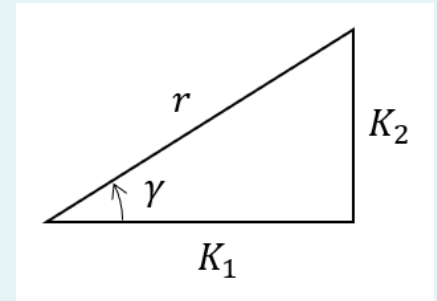
Algebraic Solution (10)

- Therefore, x and y become:

$$\begin{aligned}x &= K_1 c_1 - K_2 s_1 \\y &= K_1 s_1 + K_2 c_1\end{aligned}$$

- We will now use the **trigonometric substitution** technique.
- Introduce:

$$\begin{aligned}r &= +\sqrt{K_1^2 + K_2^2} \\ \gamma &= \text{atan2}(K_2, K_1)\end{aligned}$$

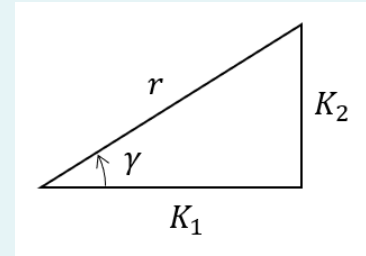


- Note: The values for both terms are known.

Algebraic Solution (11)

- With the newly introduced terms, K_1 and K_2 can be expressed as:

$$\begin{aligned} K_1 &= r \cos \gamma \\ K_2 &= r \sin \gamma \end{aligned}$$



- With these, x and y can be written as:

$$\begin{aligned} x &= K_1 c_1 - K_2 s_1 = r \cos \gamma c_1 - r \sin \gamma s_1 \\ y &= K_1 s_1 + K_2 c_1 = r \cos \gamma s_1 + r \sin \gamma c_1 \end{aligned}$$

- Or:

$$\begin{aligned} \frac{x}{r} &= \cos \gamma c_1 - \sin \gamma s_1 = \cos(\gamma + \theta_1) \\ \frac{y}{r} &= \cos \gamma s_1 + \sin \gamma c_1 = \sin(\gamma + \theta_1) \end{aligned}$$

Algebraic Solution (12)

- Finally:

$$\gamma + \theta_1 = \text{atan2}\left(\frac{y}{r}, \frac{x}{r}\right)$$

- Or:

$$\begin{aligned}\theta_1 &= \text{atan2}\left(\frac{y}{r}, \frac{x}{r}\right) - \gamma \\ &= \text{atan2}\left(\frac{y}{r}, \frac{x}{r}\right) - \text{atan2}(K_2, K_1)\end{aligned}$$

Algebraic Solution (13)

- As for θ_3 ,

$$\begin{aligned} c\phi &= c\theta_{123} \\ s\phi &= s\theta_{123} \end{aligned}$$

- Thus:

$$\theta_1 + \theta_2 + \theta_3 = \text{atan2}(s\phi, c\phi)$$

- Giving:

$$\theta_3 = \text{atan2}(s\phi, c\phi) - \theta_1 - \theta_2$$

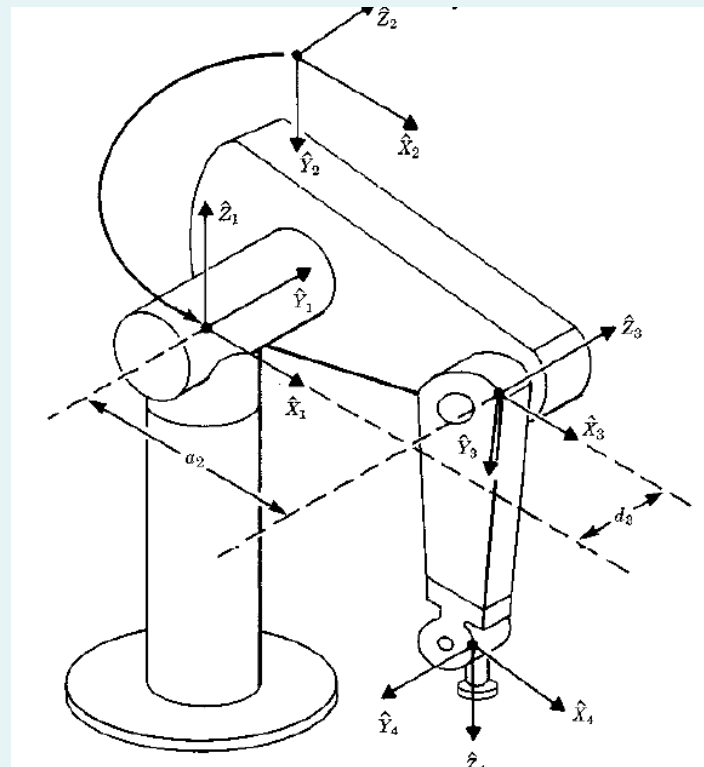
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PUMA 560 Manipulator (1)

- The link parameters of PUMA 560 Manipulator are given in the following table:

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	-90	0	0	θ_2
3	0	a_2	d_3	θ_3
4	-90	a_3	d_4	θ_4
5	90	0	0	θ_5
6	-90	0	0	θ_6



PUMA 560 Manipulator (2)

- From the DH table, first calculate the following matrices:

$${}^0_1T(\theta_1), {}^1_2T(\theta_2), {}^2_3T(\theta_3), {}^3_4T(\theta_4), {}^4_5T(\theta_5), {}^5_6T(\theta_6)$$

$${}^1_6T = {}^1_2T(\theta_2) \cdot {}^2_3T(\theta_3) \cdot {}^3_4T(\theta_4) \cdot {}^4_5T(\theta_5) \cdot {}^5_6T(\theta_6)$$

$$\vdots$$

$${}^4_6T = {}^4_5T(\theta_5) \cdot {}^5_6T(\theta_6)$$

$${}^0_1T^{-1}, {}^0_2T^{-1}, {}^0_3T^{-1}, {}^0_4T^{-1}, {}^0_5T^{-1}$$

PUMA 560 Manipulator (3)

- The last frame is required to be at:

$${}^0T_{\text{desired}} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Calculate the joint angles θ_i .

PUMA 560 Manipulator (4)

- First step: Equate desired values with general expressions:

$${}^0_6T^{\text{desired}} = {}^0_1T(\theta_1) \cdot {}^1_2T(\theta_2) \cdot {}^2_3T(\theta_3) \cdot {}^3_4T(\theta_4) \cdot {}^4_5T(\theta_5) \cdot {}^5_6T(\theta_6)$$

- Next, bring θ_1 to the left hand side:

$$[{}^0_1T(\theta_1)]^{-1} \cdot {}^0_6T^{\text{desired}} = {}^1_2T(\theta_2) \cdot {}^2_3T(\theta_3) \cdot {}^3_4T(\theta_4) \cdot {}^4_5T(\theta_5) \cdot {}^5_6T(\theta_6) = {}^1_6T$$

PUMA 560 Manipulator (5)

- This gives:

$$\begin{bmatrix} c_1 & s_1 & 0 & 0 \\ -s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = {}^1_6T = \begin{bmatrix} * & * & * & a_2c_2 + a_3c_{23} - d_4s_{23} \\ * & * & * & d_3 \\ * & * & * & -a_2s_2 - a_3s_{23} - d_4c_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Equating the (2,4) element gives:

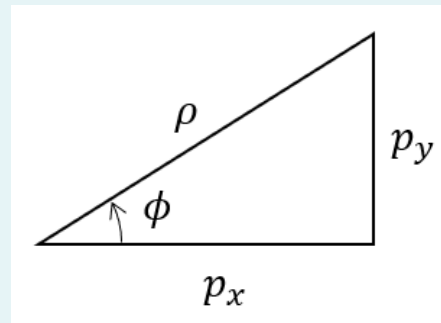
$$-s_1p_x + c_1p_y = d_3 \text{ (known constant)}$$

- Solve for θ_1 using the **trigonometric substitution** technique.

PUMA 560 Manipulator (6)

- Introduce:

$$\rho = +\sqrt{p_x^2 + p_y^2}$$
$$\phi = \text{atan2}(p_y, p_x)$$



- These give:

$$p_x = \rho \cos \phi$$
$$p_y = \rho \sin \phi$$

- Substitute into $-s_1 p_x + c_1 p_y = d_3$ gives:

$$-s_1 \cos \phi + c_1 \sin \phi = \frac{d_3}{\rho}$$

PUMA 560 Manipulator (7)

- The last equation can be written as:

$$-s_1 \cos \phi + c_1 \sin \phi = \sin(\phi - \theta_1) = \frac{d_3}{\rho}$$

- Therefore:

$$\cos(\phi - \theta_1) = \pm \sqrt{1 - \frac{d_3^2}{\rho^2}}$$

- Giving:

$$\phi - \theta_1 = \text{atan2} \left(\frac{d_3}{\rho}, \pm \sqrt{1 - \frac{d_3^2}{\rho^2}} \right)$$

PUMA 560 Manipulator (8)

- The solution for θ_1 is finally found:

$$\begin{aligned}\theta_1 &= \phi - \text{atan2}\left(\frac{d_3}{\rho}, \pm \sqrt{1 - \frac{d_3^2}{\rho^2}}\right) \\ &= \text{atan2}(p_y, p_x) - \text{atan2}\left(d_3, \pm \sqrt{p_x^2 + p_y^2 - d_3^2}\right)\end{aligned}$$

- Note: There are **two** possible answers.

PUMA 560 Manipulator (9)

- From the same expression earlier,

$$\begin{bmatrix} c_1 & s_1 & 0 & 0 \\ -s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = {}^1_6T = \begin{bmatrix} * & * & * & a_2c_2 + a_3c_{23} - d_4s_{23} \\ * & * & * & d_3 \\ * & * & * & -a_2s_2 - a_3s_{23} - d_4c_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Equating the (1,4), (2,4) and (3,4) elements gives:

$$\begin{aligned} c_1p_x + s_1p_y &= a_2c_2 + a_3c_{23} - d_4s_{23} \\ -s_1p_x + c_1p_y &= d_3 \\ -p_z &= a_2s_2 + a_3s_{23} + d_4c_{23} \end{aligned}$$

PUMA 560 Manipulator (10)

- Applying the **square and sum** technique:

$$p_x^2 + p_y^2 + p_z^2 = a_2^2 + a_3^2 + d_3^2 + d_4^2 + 2a_2a_3c_3 - 2d_4a_2s_3$$

- Or:

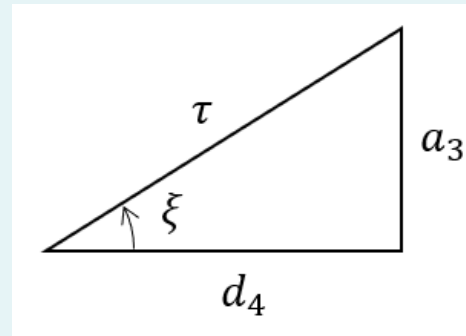
$$a_3c_3 - d_4s_3 = \frac{p_x^2 + p_y^2 + p_z^2 - a_2^2 - a_3^2 - d_3^2 - d_4^2}{2a_2} = K(\text{known constant})$$

- We will again use the **trigonometric substitution** technique to get θ_3 .

PUMA 560 Manipulator (11)

- Introduce:

$$\tau = +\sqrt{a_3^2 + d_4^2}$$
$$\xi = \text{atan2}(a_3, d_4)$$



- This means:

$$a_3 = \tau \sin \xi$$
$$d_4 = \tau \cos \xi$$

- Substitute into $a_3 c_3 - d_4 s_3 = K$ gives:

$$\tau \sin \xi c_3 - \tau \cos \xi s_3 = K$$

PUMA 560 Manipulator (12)

- The last equation gives:

$$\tau \sin \xi c_3 - \tau \cos \xi s_3 = \tau \sin(\xi - \theta_3) = K$$

- θ_3 can now be calculated from:

$$\sin(\xi - \theta_3) = \frac{K}{\tau}$$

$$\cos(\xi - \theta_3) = \pm \sqrt{1 - \frac{K^2}{\tau^2}}$$

- Giving:

$$\xi - \theta_3 = \text{artan2} \left(\frac{K}{\tau}, \pm \sqrt{1 - \frac{K^2}{\tau^2}} \right)$$

$$\theta_3 = \xi = \text{artan2}(a_3, d_4) - \text{artan2} \left(K, \pm \sqrt{a_3^2 + d_4^2 - K^2} \right)$$

- 2 solutions.

PUMA 560 Manipulator (13)

- θ_1 and θ_3 are known now. Next we will try to calculate θ_2 .
- From:

$${}^0_6T^{\text{desired}} = {}^0_1T(\theta_1) \cdot {}^1_2T(\theta_2) \cdot {}^2_3T(\theta_3) \cdot {}^3_4T(\theta_4) \cdot {}^4_5T(\theta_5) \cdot {}^5_6T(\theta_6)$$

- We can write:

$$[{}^0_1T(\theta_1) \cdot {}^1_2T(\theta_2) \cdot {}^2_3T(\theta_3)]^{-1} \cdot {}^0_6T^{\text{desired}} = {}^3_4T(\theta_4) \cdot {}^4_5T(\theta_5) \cdot {}^5_6T(\theta_6)$$

- Or:

$${}^0_3T^{-1} \cdot {}^0_6T^{\text{desired}} = {}^3_6T$$

PUMA 560 Manipulator (14)

- The last equation gives:

$$\begin{bmatrix} c_1 c_{23} & s_1 c_{23} & -s_{23} & -a_2 c_3 \\ -c_1 s_{23} & -s_1 s_{23} & -c_{23} & a_2 s_3 \\ -s_1 & c_1 & 0 & -d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} * & * & -c_4 s_5 & a_3 \\ * & * & * & d_4 \\ * & * & s_4 s_5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Equating the (1,4) and (2,4) elements from both sides:

$$\begin{aligned} c_1 c_{23} p_x + s_1 c_{23} p_y - s_{23} p_z - a_2 c_3 &= a_3 \\ -c_1 s_{23} p_x - s_1 s_{23} p_y - c_{23} p_z + a_2 s_3 &= d_4 \end{aligned}$$

PUMA 560 Manipulator (15)

- Remember that only θ_2 is unknown. Therefore:

$$\begin{aligned}c_{23}(c_1p_x + s_1p_y) - s_{23}p_z &= a_3 + a_2c_3 \\ -s_{23}(c_1p_x + s_1p_y) - c_{23}p_z &= d_4 - a_2s_3\end{aligned}$$

- We can solve these by **trigonometric substitution**. The final answer is (please try this yourselves):

$$\begin{aligned}\theta_2 &= \text{artan2}(c_1p_x + s_1p_y, p_z) \\ &\quad - \text{artan2}\left(\frac{a_3 + a_2c_3}{\sqrt{(c_1p_x + s_1p_y)^2 + p_z^2}}, \frac{a_2s_3 - d_4}{\sqrt{(c_1p_x + s_1p_y)^2 + p_z^2}}\right) - \theta_3\end{aligned}$$

- 4 solutions for 4 possible +/- combinations of θ_1 and θ_3 .

PUMA 560 Manipulator (16)

- Again from:

$$\begin{bmatrix} c_1 c_{23} & s_1 c_{23} & -s_{23} & -a_2 c_3 \\ -c_1 s_{23} & -s_1 s_{23} & -c_{23} & a_2 s_3 \\ -s_1 & c_1 & 0 & -d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} * & * & -c_4 s_5 & a_3 \\ * & * & * & d_4 \\ * & * & s_4 s_5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- we equate the (1,3) and (3,3) elements:

$$\begin{aligned} c_1 c_{23} r_{13} + s_1 c_{23} r_{23} - s_{23} r_{33} &= -c_4 s_5 \\ -s_1 r_{13} + c_1 r_{23} &= s_4 s_5 \end{aligned}$$

PUMA 560 Manipulator (17)

- If $s_5 \neq 0$, we can write:

$$c_4 = \frac{-c_1 c_{23} r_{13} - s_1 c_{23} r_{23} + s_{23} r_{33}}{s_5}$$
$$s_4 = \frac{-s_1 r_{13} + c_1 r_{23}}{s_5}$$

- This gives:

$$\theta_4 = \text{atan2}(-s_1 r_{13} + c_1 r_{23}, -c_1 c_{23} r_{13} - s_1 c_{23} r_{23} + s_{23} r_{33})$$

- 4 solutions for 4 possible +/- combinations of θ_1 and θ_3 . (θ_2 is not independent.)

PUMA 560 Manipulator (18)

- If $s_5 = 0$, then axes 4 and 6 line up, and joint 4 and joint 6 adds up to the final orientation.
 - In this case, we can only solve for $(\theta_4 + \theta_6)$ or $(\theta_4 - \theta_6)$.
 - Choose θ_4 arbitrarily and then compute θ_6 later.

PUMA 560 Manipulator (19)

- We have calculated θ_1 to θ_4 by now.
- The final steps are to solve for θ_5 and θ_6 .
- From:

$${}^0_6T^{\text{desired}} = {}^0_1T(\theta_1) \cdot {}^1_2T(\theta_2) \cdot {}^2_3T(\theta_3) \cdot {}^3_4T(\theta_4) \cdot {}^4_5T(\theta_5) \cdot {}^5_6T(\theta_6)$$

- We can write:

$$[{}^0_1T(\theta_1) \cdot {}^1_2T(\theta_2) \cdot {}^2_3T(\theta_3) \cdot {}^3_4T(\theta_4)]^{-1} \cdot {}^0_6T^{\text{desired}} = {}^4_5T(\theta_5) \cdot {}^5_6T(\theta_6)$$

- Or:

$${}^0_4T^{-1} \cdot {}^0_6T^{\text{desired}} = {}^4_6T$$

PUMA 560 Manipulator (20)

- The last equation gives:

$$\begin{bmatrix} c_1 c_{23} c_4 + s_1 s_4 & s_1 c_{23} c_4 - c_1 s_4 & -s_{23} c_4 & -a_2 c_3 c_4 + d_3 s_4 - a_3 c_4 \\ -c_1 c_{23} s_4 + s_1 c_4 & -s_1 c_{23} s_4 - c_1 c_4 & s_{23} s_4 & a_2 c_3 s_4 + d_3 c_4 + a_3 s_4 \\ -c_1 s_{23} & -s_1 s_{23} & -c_{23} & a_2 s_3 - d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} * & * & -s_5 & 0 \\ * & * & * & 0 \\ * & * & c_5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

PUMA 560 Manipulator (21)

- Equating the (1,3) and (3,3) elements, we have:

$$\begin{aligned} -(c_1 c_{23} c_4 + s_1 s_4) r_{13} - (s_1 c_{23} c_4 - c_1 s_4) r_{23} + s_{23} c_4 r_{33} &= s_5 \\ -(c_1 s_{23}) r_{13} + (-s_1 s_{23}) r_{23} - c_{23} r_{33} &= c_5 \end{aligned}$$

- We can therefore solve for θ_5 easily:

$$\theta_5 = \text{atan2}(s_5, c_5)$$

- 4 solutions for 4 possible +/- combinations of θ_1 and θ_3 . (θ_2 and θ_4 are not independent.)

PUMA 560 Manipulator (22)

- Applying the same method again:

$${}^0T^{-1} \cdot {}^0T^{\text{desired}} = {}^5T$$

$$\begin{bmatrix} (c_1 c_{23} c_4 + s_1 s_4) c_5 - c_1 s_{23} s_5 & (s_1 c_{23} c_4 - c_1 s_4) c_5 - s_1 s_{23} s_5 & -s_{23} c_4 c_5 & * \\ * & * & * & * \\ c_1 c_{23} s_4 - s_1 c_4 & s_1 c_{23} s_4 + c_1 c_4 & -s_{23} s_4 & * \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c_6 & -s_6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s_6 & -c_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

PUMA 560 Manipulator (23)

- The (1,3) and (3,3) elements are:

$$\begin{aligned} & ((c_1 c_{23} c_4 + s_1 s_4) c_5 - c_1 s_{23} s_5) r_{11} + ((s_1 c_{23} c_4 - c_1 s_4) c_5 \\ & - s_1 s_{23} s_5) r_{21} + (-s_{23} c_4 c_5) r_{31} = c_6 \\ & -(c_1 c_{23} s_4 - s_1 c_4) r_{11} - (s_1 c_{23} s_4 + c_1 c_4) r_{21} + (s_{23} s_4) r_{31} = s_6 \end{aligned}$$

- Therefore:

$$\theta_6 = \text{atan2}(s_6, c_6)$$

- 4 solutions for 4 possible +/- combinations of θ_1 and θ_3 . (θ_2 , θ_4 and θ_5 are not independent.)

PUMA 560 Manipulator (24)

- Final note: Apart from the four solutions to θ 's computed above, there are four more solutions obtained by flipping the wrist of manipulator, via:

$$\theta'_4 = \theta_4 + 180$$

$$\theta'_5 = -\theta_5$$

$$\theta'_6 = \theta_6 + 180$$

General Notes

- The example of PUMA 560 showed one useful technique:
 - Isolate one joint variable at a time.

$$[{}^0_1T(q_1)]^{-1} \cdot {}^0_nT^{\text{desired}} = {}^1_2T(q_2) \cdot {}^2_3T(q_3) \cdot \dots \cdot {}^{n-2}_{n-1}T(q_{n-1}) \cdot {}^{n-1}_nT(q_n) = {}^1_nT$$

- Look for constant elements in 1_nT .
- Equate $\text{LHS}(i,j) = \text{RHS}(i,j)$, and then solve for q_1 .
- After this, isolate the next variable:

$$[{}^1_2T(q_2)]^{-1} \cdot [{}^0_1T(q_1)]^{-1} \cdot {}^0_nT^{\text{desired}} = {}^2_3T(q_3) \cdot \dots \cdot {}^{n-2}_{n-1}T(q_{n-1}) \cdot {}^{n-1}_nT(q_n) = {}^2_nT$$

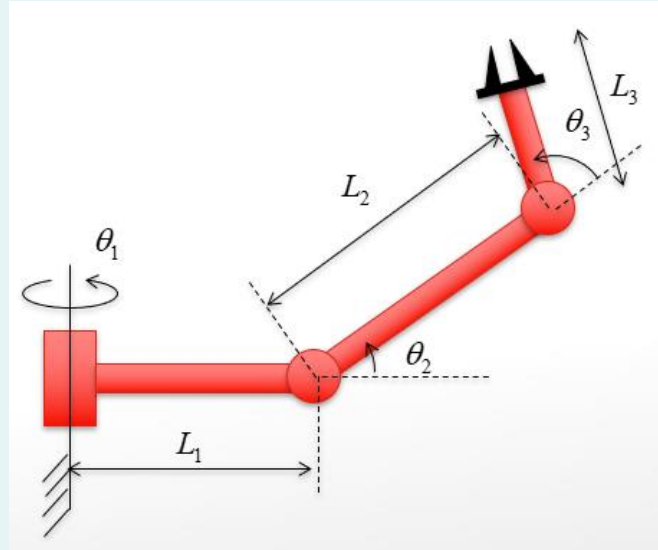
- Look for constant elements in 2_nT .
- Equate $\text{LHS}(i,j) = \text{RHS}(i,j)$, and then solve for q_2 .
- Repeat until q_n .

Content

- Introduction to Inverse Kinematics
- Existence and Uniqueness of Solution
- Geometric Solution
- Algebraic Solution
- Another Example
- Tutorial Questions

Question 1

- Sketch the workspace of the following manipulator with:
 - $L_1 = 15$
 - $L_2 = 10$
 - $L_3 = 3$

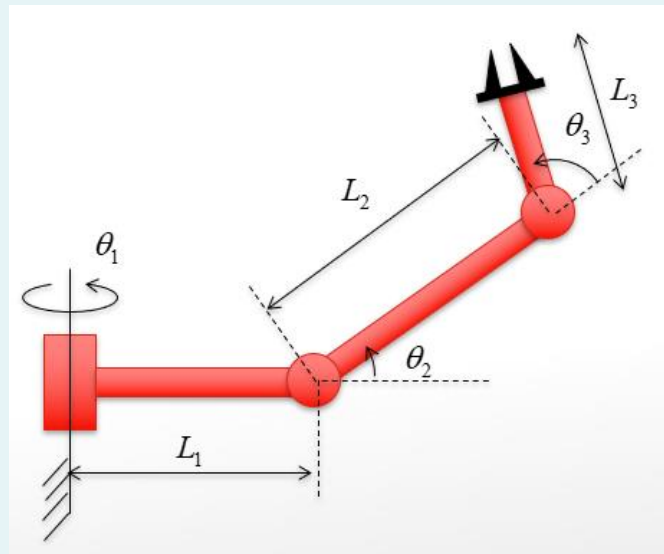


Question 2

- The last frame of the 3-link manipulator on the right is required to be at:

$$\begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- What should the joint angles be?

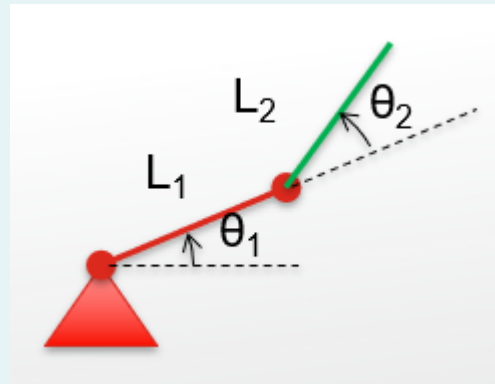


Question 3

- Given a desired position and orientation of the hand of a three-link planar rotary jointed manipulator,
 - there are two possible solutions.
 - If one more rotational joints is added (while the arm is still planar),
 - how many solutions are there?

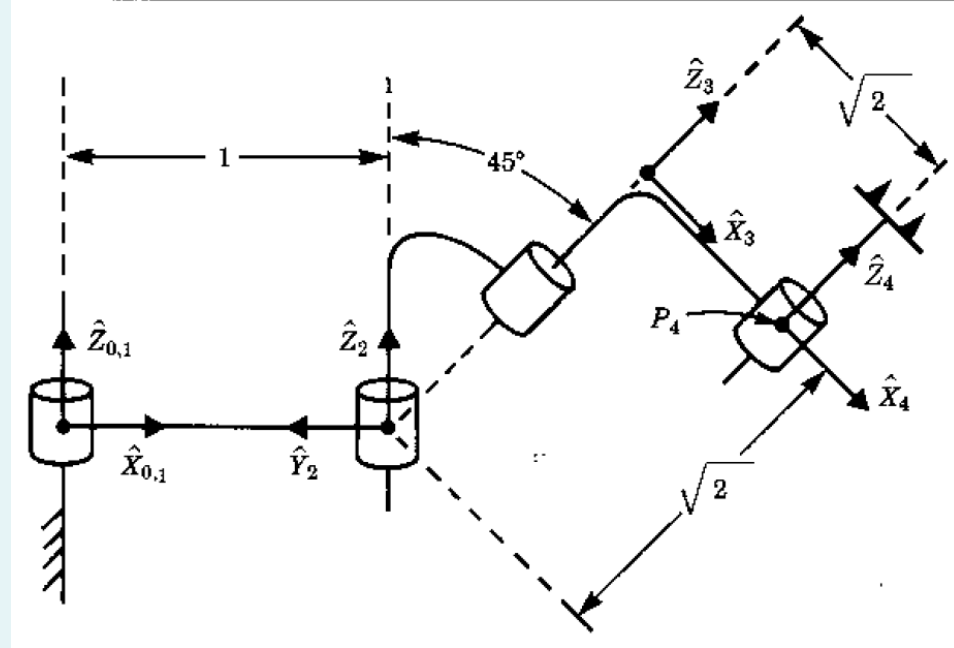
Question 4

- For the two-link robot as shown, the second link is half as long as the first ($L_1 = 2L_2$).
 - The joint range limits are:
$$0 < \theta_1 < 180$$
$$-90 < \theta_2 < 180$$
- Sketch the approximate reachable workspace of the tip of link 2.



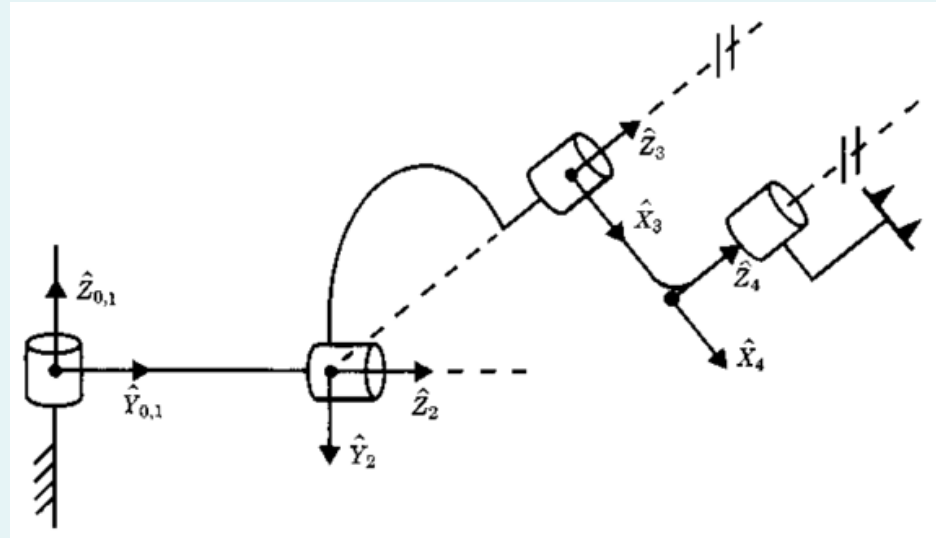
Question 5

- For the 4R manipulator as shown, the non-zero link parameters are:
 $a_1 = 1, \alpha_2 = 45^\circ, d_3 = \sqrt{2}, a_3 = \sqrt{2}$
- The configuration as shown is:
 $\theta_1 = 0, \theta_2 = 90^\circ, \theta_3 = -90^\circ, \theta_4 = 0^\circ$
- Each joint has limit of $\pm 180^\circ$.
- Find all values of θ_3 such that:
 ${}^0P_{4ORG} = [1.1 \quad 1.5 \quad 1.707]^T$.



Question 6

- For the 4R manipulator as shown, the non-zero link parameters are: $\alpha_1 = -90^\circ$, $\alpha_2 = 45^\circ$, $d_2 = 1$, $d_3 = 1$, $a_3 = 1$.
- The configuration as shown is: $\theta_1 = 0$, $\theta_2 = 0^\circ$, $\theta_3 = 90^\circ$, $\theta_4 = 0^\circ$
- Each joint has limit of $\pm 180^\circ$.
- Find all values of θ_3 such that:
$${}^0P_{4ORG} = \begin{bmatrix} 0 & 1 & 1.414 \end{bmatrix}^T.$$



Thank you for your attention!

Any questions?